Homework Practice Sheet (Intro ML / CNNs)
Convolutions, Output Shapes + Concepts

Instructions

Show your work. Unless otherwise stated, use the CNN convention for convolution layers (*no kernel flipping*; i.e. cross-correlation as used in practice).

For a convolution with input height/width $H \times W$, kernel size K, padding P, and stride S, use:

$$H_{\text{out}} = \left\lfloor \frac{H + 2P - K}{S} \right\rfloor + 1, \qquad W_{\text{out}} = \left\lfloor \frac{W + 2P - K}{S} \right\rfloor + 1.$$

For conceptual questions, answer in 1–3 sentences.

Part A: Numeric Problems (CNN Basics)

1. **Convolution output dimensions.**
   An input image has shape $8 \times 8 \times 1$. A convolution layer uses:

   $$\text{kernel size } 3 \times 3, \qquad \text{stride } 1, \qquad \text{padding } 0,$$

   with **4 filters**. Compute the output shape.

2. **Convolution output dimensions with padding.**
   An input image has shape $10 \times 10 \times 1$. A convolution layer uses:

   $$\text{kernel size } 3 \times 3, \qquad \text{stride } 1, \qquad \text{padding } 1,$$

   with **2 filters**. Compute the output shape.

3. **Number of parameters in a convolution layer.**
   A convolution layer takes an input with **1 channel** and uses **3 filters**, each of size 3×3. Assume each filter has its own bias term.
   How many trainable parameters are in this layer?

4. **Small convolution by hand.**
   Let the input feature map be

   $$X = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$$

   and let the kernel be

   $$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

   Using stride 1 and padding 0, compute the full output feature map.

5. **Max-pooling.**
   Apply 2×2 max-pooling with stride 2 to

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 4 & 2 & 5 & 1 \\ 0 & 2 & 7 & 3 \\ 1 & 6 & 4 & 2 \end{bmatrix}.$$

   Compute the output.

Part B: Conceptual Questions (Short Answers)

6. **Why do CNNs use filters?**
   In simple terms, what is a filter in a CNN trying to detect?

7. **What does padding do?**
   What is the purpose of padding in a convolutional layer?

8. **What does max-pooling do?**
   Why might we use max-pooling in a CNN?

1. $H_{out} = \left\lfloor \frac{8 + 2\times 0 - 3}{1} \right\rfloor + 1$

$= \left\lfloor \frac{8-3}{1} \right\rfloor + 1$

$= 6$

$W_{out} = \left\lfloor \frac{8 + 2\times 0 - 3}{1} \right\rfloor + 1$

$= 6$

Since there are 4 filters, the output shape is $6\times 6\times 4$

2. $H_{out} = \left\lfloor \frac{10 + 2\times 1 - 3}{1} \right\rfloor + 1$

$= \left\lfloor \frac{12-3}{1} \right\rfloor + 1$

$= 10$

Similarly, $W_{out} = 10$

Dimensions are $10\times 10\times 2$

3. $3\times 3 + 1 = 10$

$10\times 3 = 30$ parameters

4. $X = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$

$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$1\times 1 + 2\times 0 + 0\times 0 + 1\times 1 = 2$

$2\times 1 + 0\times 0 + 1\times 0 + 1\times 1 = 3$

$0\times 1 + 1\times 0 + 2\times 0 + 1\times 1 = 1$

$1\times 1 + 0\times 1 + 0\times 1 + 1\times 0 = 1$

$\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}$

5. $\begin{bmatrix} 1 & 3 & 2 & 0 \\ 4 & 2 & 5 & 1 \\ 0 & 2 & 7 & 3 \\ 1 & 6 & 4 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$

6. we use filters to detect a local pattern in the input. when we slide the kernal across the image, we figure out where the feature appears.

7. we add extra values around the border of the input, preserving the spatial dimensions of the output.

8. Max-pooling takes the max value in local window. useful for reducing the spatial dimensions and number of computations picks out most important features.